

✓ Homework 8: SQL

After going through the lecture notebooks, work on the homework below.

Question

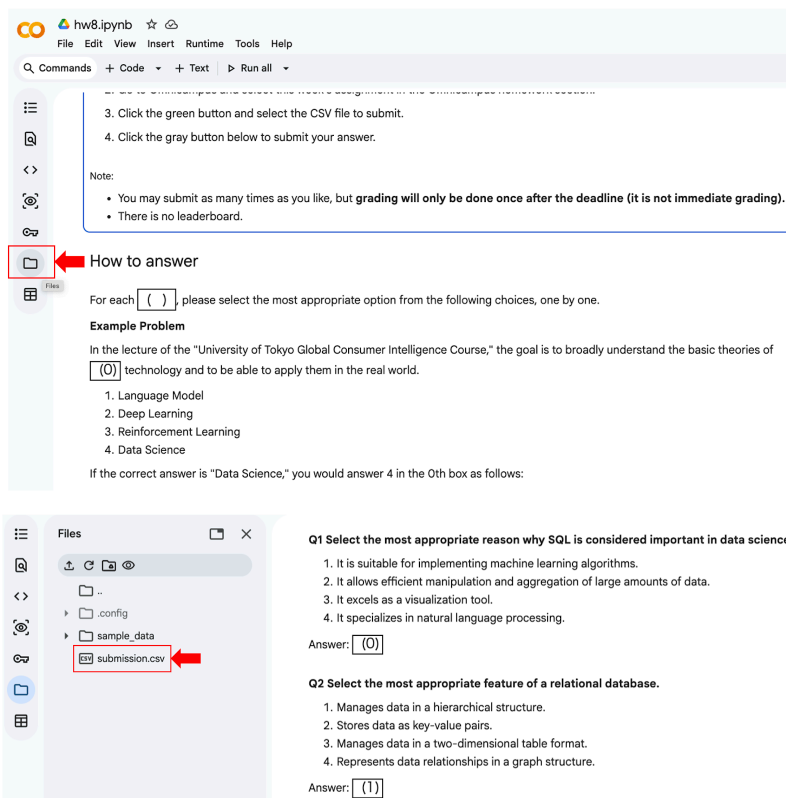
This homework is quiz-style assignment.

Choose the answer you think is correct, then write your answer in the code block below. Run all code blocks to generate `submission.csv`. Download the file, then submit to Omnicampus.

A score of 8 or more out of 10 is considered correct.

Submission Guidelines:

1. Download your `submission.csv` from the folder icon on the left. Click the  button or right click the file to download.



The screenshot shows a Jupyter Notebook interface. On the left, there is a file explorer sidebar. A red arrow points to a folder icon labeled 'Files'. Below it, a file named 'submission.csv' is highlighted with a red box and a red arrow. The main area of the notebook displays a quiz question titled 'How to answer'. The question asks to select the most appropriate option from a list of choices for a given scenario. The choices are: 1. Language Model, 2. Deep Learning, 3. Reinforcement Learning, 4. Data Science. The question also includes an example problem and a note about grading. Below the question, there are two more questions, Q1 and Q2, each with a list of options and a text input field for the answer.

How to answer

For each `()`, please select the most appropriate option from the following choices, one by one.

Example Problem

In the lecture of the "University of Tokyo Global Consumer Intelligence Course," the goal is to broadly understand the basic theories of `()` technology and to be able to apply them in the real world.

1. Language Model
2. Deep Learning
3. Reinforcement Learning
4. Data Science

If the correct answer is "Data Science," you would answer 4 in the 0th box as follows:

Q1 Select the most appropriate reason why SQL is considered important in data science.

1. It is suitable for implementing machine learning algorithms.
2. It allows efficient manipulation and aggregation of large amounts of data.
3. It excels as a visualization tool.
4. It specializes in natural language processing.

Answer: `()`

Q2 Select the most appropriate feature of a relational database.

1. Manages data in a hierarchical structure.
2. Stores data as key-value pairs.
3. Manages data in a two-dimensional table format.
4. Represents data relationships in a graph structure.

Answer: `()`

2. Go to Omnicampus and select this week's assignment in the Omnicampus homework section.
3. Click the green button and select the CSV file to submit.

4. Click the gray button below to submit your answer.

Note:

- You may submit as many times as you like, but **grading will only be done once after the deadline (it is not immediate grading)**.
- There is no leaderboard.

✓ How to answer

For each , please select the most appropriate option from the following choices, one by one.

Example Problem

In the lecture of the "University of Tokyo Global Consumer Intelligence Course," the goal is to broadly understand the basic theories of technology and to be able to apply them in the real world.

1. Language Model
2. Deep Learning
3. Reinforcement Learning
4. Data Science

If the correct answer is "Data Science," you would answer 4 in the 0th box as follows:

```
example[0] = 4
```

Deadline

Wed, Jun 17 (11:00 UTC)

✓ Quiz

Q1 Select the most appropriate reason why SQL is considered important in data science.

1. It is suitable for implementing machine learning algorithms.
2. It allows efficient manipulation and aggregation of large amounts of data.
3. It excels as a visualization tool.
4. It specializes in natural language processing.

Answer:

Q2 Select the most appropriate feature of a relational database.

1. Manages data in a hierarchical structure.
2. Stores data as key-value pairs.
3. Manages data in a two-dimensional table format.
4. Represents data relationships in a graph structure.

Answer:

Q3 Select the most appropriate role of a "primary key" in a database.

1. Data encryption
2. Unique identification of records
3. Joining tables
4. Data compression

Answer:

Q4 Select the process that is not included in the ETL process.

1. Perform data cleansing and standardization.
2. Collect data from multiple data sources.
3. Execute advanced analysis using the stored data.
4. Store the transformed data in the target system.

Answer:

Q5 Select the most appropriate feature of the WHERE clause and HAVING clause in databases.

1. WHERE clause is applied to data before aggregation, and HAVING clause is applied to data after aggregation.
2. WHERE clause is applied to data after aggregation, and HAVING clause is applied to data before aggregation.

3. WHERE clause is used together with the GROUP BY clause, and HAVING clause is used alone.
4. WHERE clause is only applied to numerical data, and HAVING clause is only applied to text data.

Answer:

Q6 Select the most appropriate feature of a LEFT OUTER JOIN in databases.

1. Includes all data from the left table in the result, even if there is no matching data in the right table.
2. Includes all data from the right table in the result, even if there is no matching data in the left table.
3. Includes only data that matches in both tables in the result.
4. Includes all data from both tables in the result.

Answer:

Q7 Select the most appropriate clause that fits in the underlined part of the following SQL statement, in accordance with the (condition):

```
SELECT _____(*) FROM employees WHERE salary >= 300000;
```

(Condition) Retrieve the number of employees from the employees table with salaries of 300,000 or more.

1. SUM
2. COUNT
3. AVG
4. NUMBER

Answer:

Q8 Select the most appropriate clause that fits in the underlined part of the following SQL statement, in accordance with the (condition):

```
SELECT name, salary FROM employees ORDER BY salary _____ 5;
```

(Condition) Retrieve the names and salaries of employees from the employees table, and select the top 5 highest salaries.

1. ASC, LIMIT
2. DESC, LIMIT
3. ASC, TOP
4. DESC, TOP

Answer:

Q9 Select the most appropriate description of the execution result of the following SQL statement.

```
SELECT product_name FROM products WHERE price > (SELECT AVG(price) FROM products);
```

1. Names of the most expensive products.
2. Names of products exceeding the average price.
3. Names of products below the average price.
4. Names of products with prices equal to the average price.

Answer:

Q10 Select the most appropriate description of the execution result of the following SQL statement.

```
SELECT
    product_name,
    CASE
        WHEN in_stock = 0 THEN 'Out of Stock'
        WHEN in_stock < 10 THEN 'Low Stock'
        ELSE 'In Stock'
    END AS stock_status
FROM products;
```

1. Stock status of products with no inventory
2. Stock status of products with the highest inventory
3. Names and inventory counts of each product
4. Names and inventory status of each product

Answer:



Solution

Fill in the WRITE ME blocks with all cells to create your submission.

```
# Please do not modify this part
import numpy as np
import pandas as pd

NUM_EXAMPLE_QUESTIONS = 1
example = np.zeros(NUM_EXAMPLE_QUESTIONS, int)

NUM_QUESTIONS = 10
myanswer = np.zeros(NUM_QUESTIONS, int)
```

```
myanswer[0] = # WRITE ME
myanswer[1] = # WRITE ME
myanswer[2] = # WRITE ME
myanswer[3] = # WRITE ME
myanswer[4] = # WRITE ME
myanswer[5] = # WRITE ME
myanswer[6] = # WRITE ME
myanswer[7] = # WRITE ME
myanswer[8] = # WRITE ME
myanswer[9] = # WRITE ME
```

```
# Create CSV
if not np.any(myanswer==0):
    submission = pd.Series(myanswer, name='label')
    submission.to_csv('submission.csv', header=True, index_label='id')
else:
    print('Error: please answer all of the questions')
```

